

Experiment Documentation

1. Objective

The objective of this study is to develop a **privacy-preserving federated learning-based framework** for diabetes prediction. The proposed approach aims to enable collaborative model training across multiple distributed clients without sharing sensitive patient data. Additionally, the study evaluates whether federated learning can achieve comparable or improved performance relative to traditional centralized approaches while maintaining data privacy.

2. Dataset

The experiments are conducted using multiple datasets to ensure robustness and generalization:

- **PIMA Indian Diabetes Dataset:** A widely used benchmark dataset containing diagnostic medical attributes for diabetes prediction.
- **Diabetes Dataset:** Used to validate model generalization across different data distributions.
- **Diabetes Dataset with Clinical Notes (Multimodal):** Incorporates structured data along with textual clinical information to evaluate the effectiveness of multimodal learning.

Data preprocessing steps include handling missing values, normalization, and feature selection. For federated learning, the datasets are distributed among multiple clients to simulate real-world decentralized healthcare environments.

3. Experimental Setup

The proposed system is implemented using a **Federated Stacking Model**, where multiple base learners are combined to improve predictive performance.

- **Model Architecture:** Stacking ensemble integrated within a federated learning framework
- **Clients:** 3–5 distributed clients simulating different hospitals or data sources
- **Communication Rounds:** 10–50 rounds for model aggregation
- **Training Process:**
 - Local training is performed on each client
 - Model parameters are aggregated at the central server
 - No raw data is shared between clients
- **Additional Setup:**
 - Non-IID data distribution across clients
 - Hyperparameter tuning for optimal performance

4. Baseline

To evaluate the effectiveness of the proposed approach, a **centralized model** is implemented as a baseline.

- All data is combined and trained in a single environment
- Same model architecture (stacking) is used for fair comparison
- Performance is compared with the federated model

5. Evaluation Metrics

The performance of the models is evaluated using standard classification metrics:

- **Accuracy:** Overall correctness of predictions
- **Precision:** Measure of positive prediction correctness
- **Recall:** Ability to identify positive cases
- **F1-Score:** Harmonic mean of precision and recall
- **ROC-AUC:** Ability of the model to distinguish between classes

These metrics provide a comprehensive evaluation of model performance.

6. Results

The results are presented using both quantitative and visual analysis:

- **Performance Tables:** Comparison between federated and centralized models
- **Graphs:**
 - Accuracy per communication round
 - Loss curves
 - ROC curves
- **Observations:**
 - Federated model shows competitive performance
 - Improved generalization across distributed datasets
 - Privacy preservation without significant performance loss

7. Conclusion

The experimental results demonstrate that the proposed **federated learning-based stacking model** is effective for diabetes prediction in distributed healthcare settings.

- The federated approach achieves performance comparable to the centralized model
- It ensures **data privacy and security**, which is critical in medical applications
- The use of multiple datasets improves model robustness and generalization

However, challenges such as communication overhead and data heterogeneity still exist. Future work may focus on optimizing communication efficiency and integrating advanced explainability techniques for better clinical interpretability.